

An end-to-end quantum algorithm for nonlinear fluid dynamics with bounded quantum advantage



David Jennings, **Kamil Korzekwa**, Matteo Lostaglio, Paul Mannix



Richard Ashworth, Emanuele Marsili, Stephen Rolston

THE TEAM



David Jennings



Kamil Korzekwa



Richard Ashworth



Emanuele Marsili



Matteo Lostaglio



Paul Mannix



Stephen Rolston



AGENDA

01 Introduction

Background & summary

- Motivation
- Research landscape
- Summary of results

02 Framework

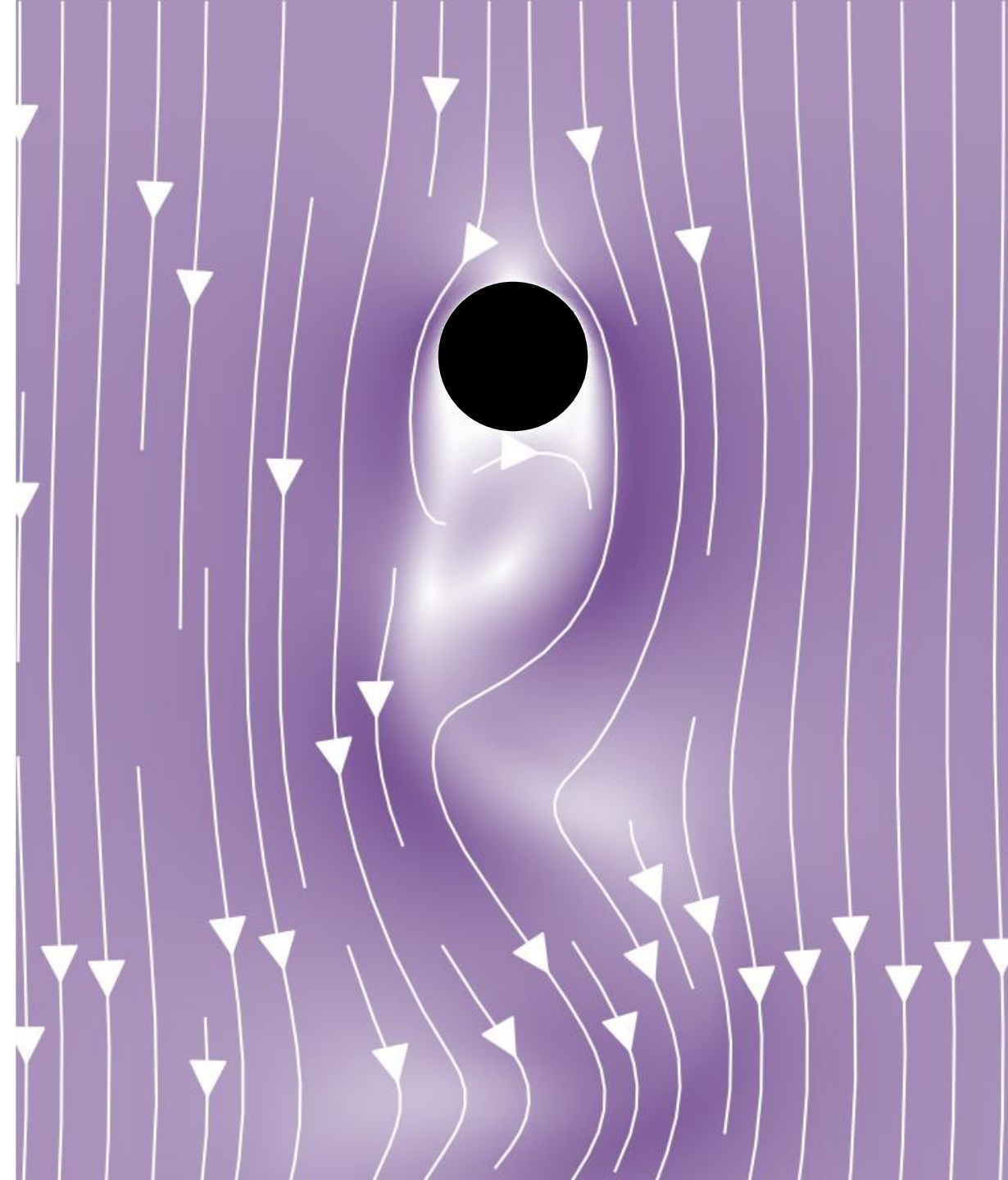
CFD problems & pipeline

03 Quantum algorithm

Performance & accuracy of the quantum LBE algorithm

04 Conclusions & outlook

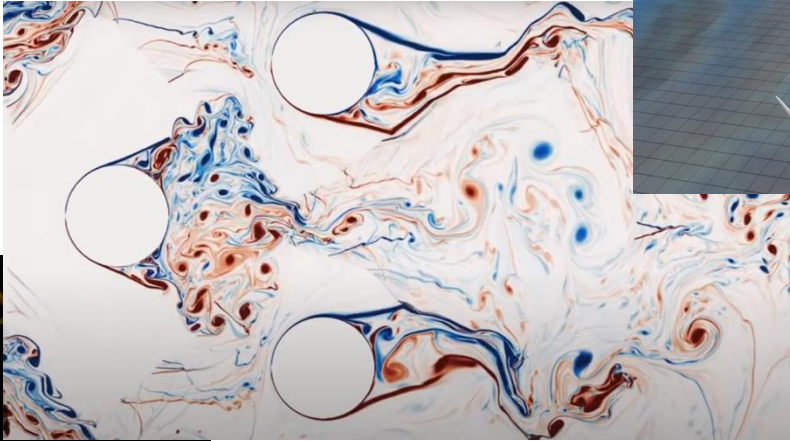
Main takeaways & areas for improvement



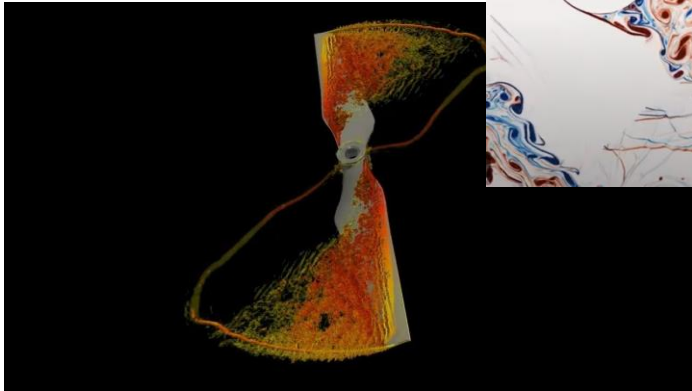
INTRODUCTION

Motivation

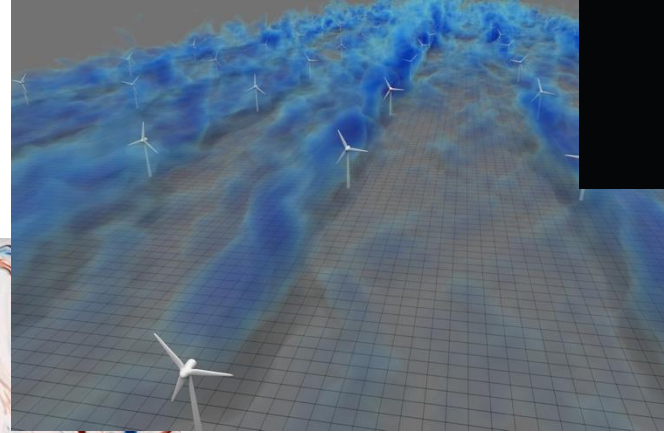
$$\text{Complexity}_C = O(\text{Re}^3)$$



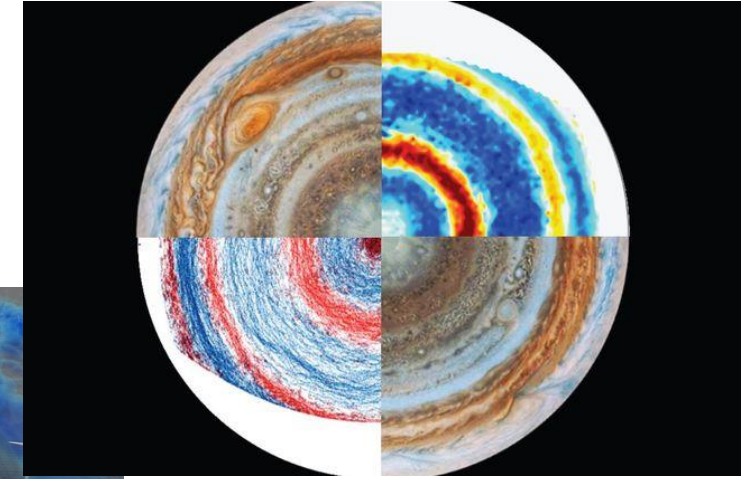
DNS - vortex shedding $\text{Re} = 2 \times 10^5$ (John Hopkins)



DNS - starting drone rotor $\text{Re} = 1.5 \times 10^4$ (FAU, KTH, KAUST)



LES - wind farm $\text{Re} = 10^8$ (John Hopkins)



Experiment – Jupiter's jets
 $\text{Re} \sim 10^{10}$ (UCLA/IRPHE)

$$\text{Complexity}_Q = ?$$

INTRODUCTION

Research landscape

	Year	Analysis	Speed-up	Resource estimates	Accuracy	CFD apps
Potential quantum advantage for simulation of fluid dynamics Xiangyu Li,^{1,*} Xiaolong Yin,² Nathan Wiebe,^{3,1,4} Jaehun Chun,^{1,5} Gregory K. Schenter,¹ Margaret S. Cheung,^{1,6} and Johannes Mülmenstädt^{1,†} ¹Pacific Northwest National Laboratory, Richland, WA, USA ²Petroleum Engineering, Colorado School of Mines, Golden, CO, USA ³Department of Computer Science, University of Toronto, Toronto, ON, Canada ⁴Canadian Institute for Advanced Research, Toronto, ON, Canada ⁵Levich Institute and Department of Chemical Engineering, CUNY City College of New York, New York, NY, USA ⁶Department of Physics, University of Washington, Seattle, WA, USA	2023	Initial ✗	None (claimed exp) ✗	Preliminary ✗	Unverified ✗	No ✗
Detailed assessment of calculating drag force with quantum computers: Explicit time-evolution precludes exponential advantage for nonlinear differential equations John Penuel¹, Amara Katabarwa^{2,3}, Peter D. Johnson³, Parker Kuklinski⁴, Benjamin Rempfer⁴, Collin Farquhar³, Yudong Cao^{3,6}, and Michael C. Garrett^{†1} ¹L3Harris Technologies, Inc., Palm Bay, FL, USA ²Georgia Tech Research Institute, Atlanta, GA, USA ³Zapata AI, Boston, MA, USA ⁴MIT Lincoln Laboratory, Lexington, MA, USA ⁶Boston Consulting Group, 200 Pier 4 Blvd, Boston, MA	2024	Flawed end-to-end ◐	Flawed modest poly ◐	Partial ◐	Unverified ✗	No ✗
An end-to-end quantum algorithm for nonlinear fluid dynamics with bounded quantum advantage David Jennings¹, Kamil Korzekwa^{*1}, Matteo Lostaglio¹, Richard Ashworth², Emanuele Marsili², and Stephen Rolston² ¹PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA ²Airbus Operations Ltd, Pegasus House Aerospace Avenue, Filton, Bristol, UK Simulating non-trivial incompressible flows with a quantum lattice Boltzmann algorithm David Jennings[*], Kamil Korzekwa^{*†}, Matteo Lostaglio[*], and Paul Mannix^{*†} <i>PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA</i> Richard Ashworth[†], Emanuele Marsili[‡], and Stephen Rolston[‡] <i>Airbus Operations Ltd, Pegasus House Aerospace Avenue, Filton, Bristol, UK</i>	2025	End-to-end ✓	Modest poly ◐	Complete ✓	Partially confirmed ◐	Yes ✓

INTRODUCTION

Summary of results

Identifying complexity bottlenecks

Lack of linearization convergence & no quantum advantage in existing proposals

Classical problem reformulation

Use incompressible LBE & discrete Carleman embedding to avoid bottlenecks

Quantum algorithm design & performance analysis

End-to-end algorithm with detailed accuracy & resource analysis (up to the gate level) exhibiting quantum advantage

Test on benchmark CFD problems

Accuracy and complexity of the algorithm tested on non-trivial CFD problems

AGENDA

0 1 Introduction

Background & summary

0 2 Framework

CFD problems & pipeline

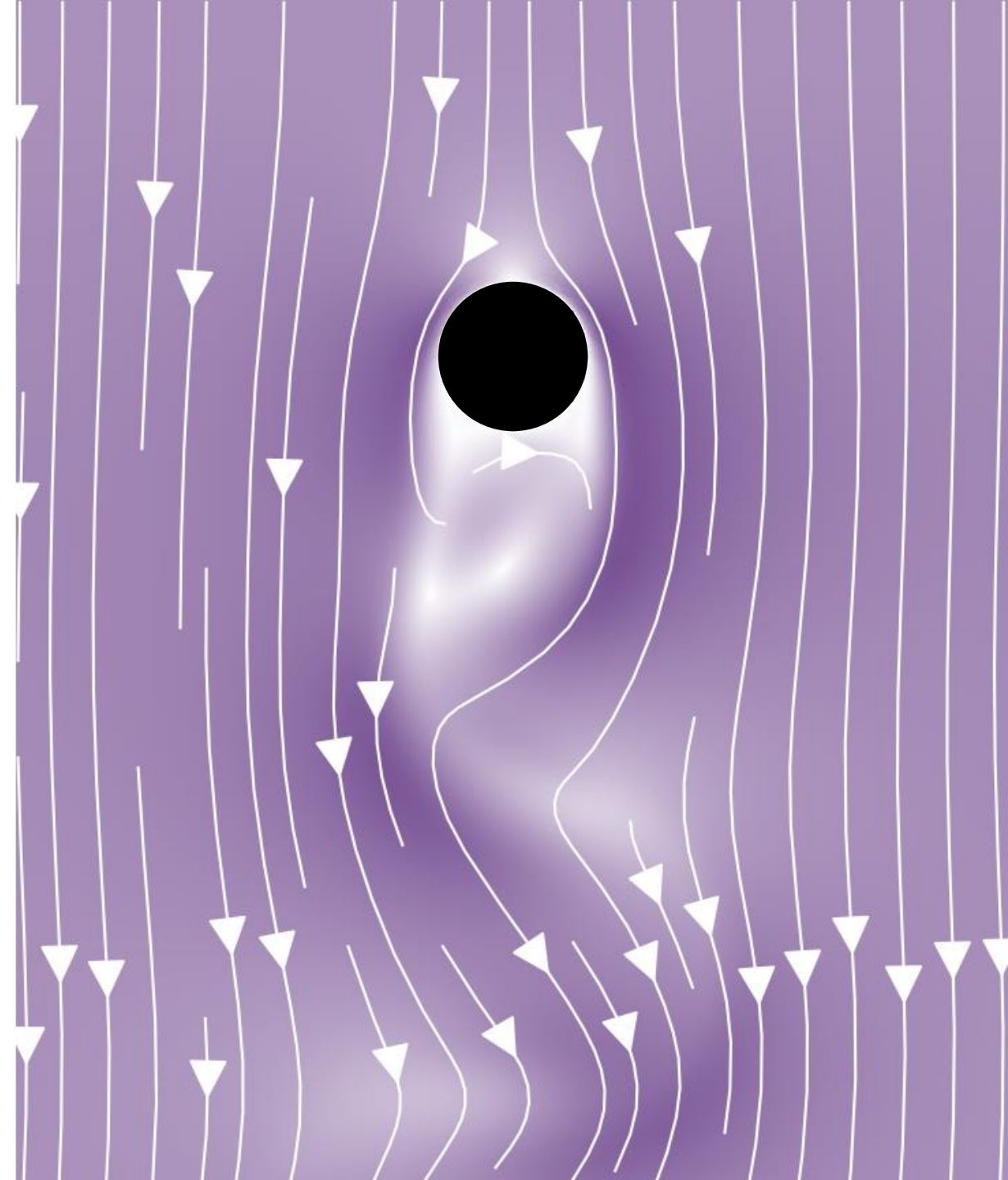
- Fluid dynamics models
- Pipeline for nonlinear differential equations
- Identifying complexity bottlenecks
- Classical problem reformulation

0 3 Quantum algorithm

Performance & accuracy of the quantum LBE algorithm

0 4 Conclusions & outlook

Main takeaways & areas for improvement



FRAMEWORK

Fluid dynamics models

Incompressible Navier-Stokes equations (nonlinearity scales with Re)

Momentum equation $\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u} + \mathbf{F}$

Mass conservation $\nabla \cdot \mathbf{u} = 0$

$\mathbf{u}$ – velocity field

p – pressure field

ν – kinematic viscosity

$\mathbf{F}$ – external force

Discrete-Boltzmann equation (nonlinearity scales with Ma)

$$\partial_t f_m(\mathbf{r}, t) + \mathbf{e}_m \cdot \nabla f_m(\mathbf{r}, t) = \Omega_m(\mathbf{r}, t)$$

$$p = c_s^2 \sum_{m=1}^Q f_m(\mathbf{r}, t)$$

$$\mathbf{u}(\mathbf{r}, t) = \frac{1}{\rho_0} \sum_{m=1}^Q f_m(\mathbf{r}, t) \cdot \mathbf{e}_m$$

$\mathbf{e}_m \cdot \nabla f_m(\mathbf{r}, t)$ – streaming term

$\Omega_m(\mathbf{r}, t)$ – collision term

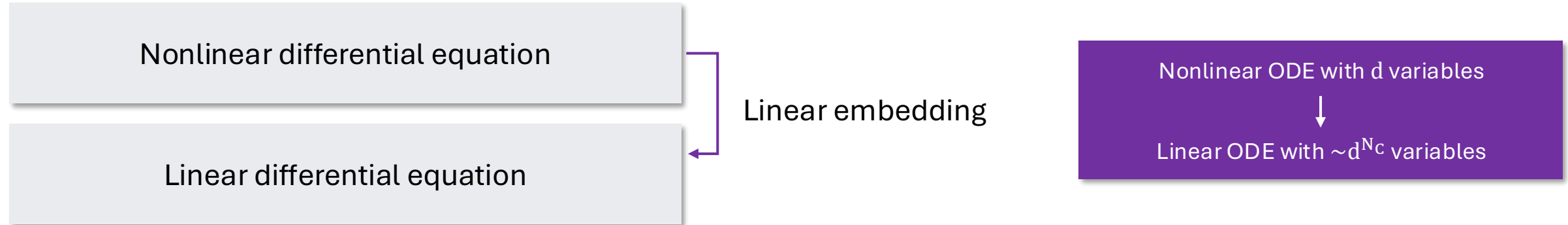
$f_m(\mathbf{r}, t)$ – discretized density of particles

$\mathbf{e}_m$ – unit velocity in direction $m \in \{1, \dots, Q\}$

$Q = 3/9/27$ – number of unit velocities available in 1D/2D/3D

FRAMEWORK

Pipeline for nonlinear differential equations



Crash course on Carleman embedding

Consider a nonlinear ODE:

$$\dot{x}_1(t) = a x_1(t) + b x_1^2(t)$$

Introduce a new variable:

$$x_2(t) := x_1^2(t)$$

Get a linear ODE for the old variable:

$$\dot{x}_1(t) = a x_1(t) + b x_2(t)$$

But a nonlinear ODE for the new variable

$$\dot{x}_2(t) = 2 a x_2(t) + 2 b x_1(t) x_2(t)$$

$$x_3(t) := x_1^3(t)$$

$$\dot{x}_2(t) = 2 a x_2(t) + 2 b x_3(t)$$

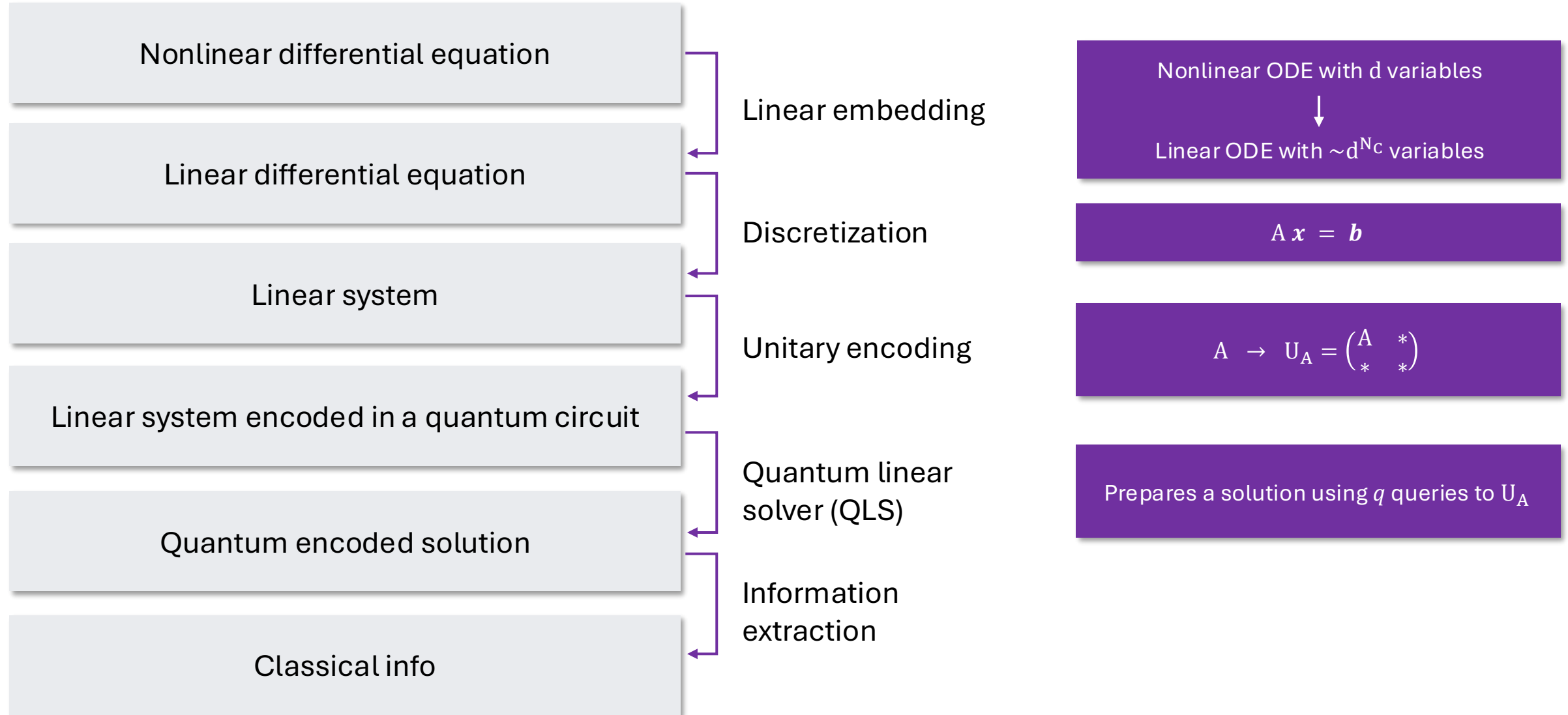
⋮

$$\dot{x}_{N_c}(t) = N_c a x_{N_c}(t) + 2 b x_1(t) x_{N_c}(t)$$

$$x \rightarrow y = [x, x^{\otimes 2}, \dots, x^{\otimes N_c}]$$

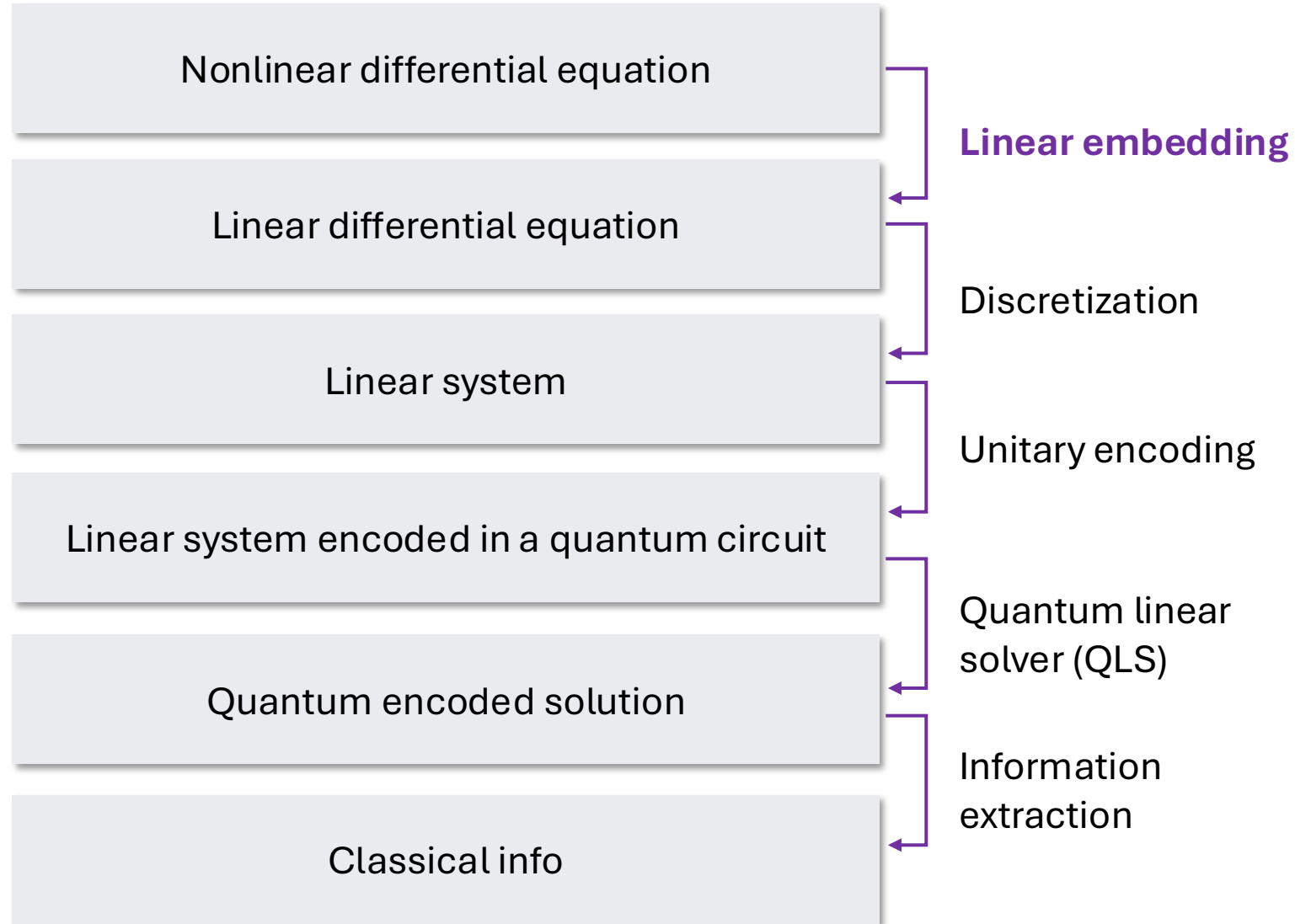
FRAMEWORK

Pipeline for nonlinear differential equations



FRAMEWORK

Complexity bottlenecks

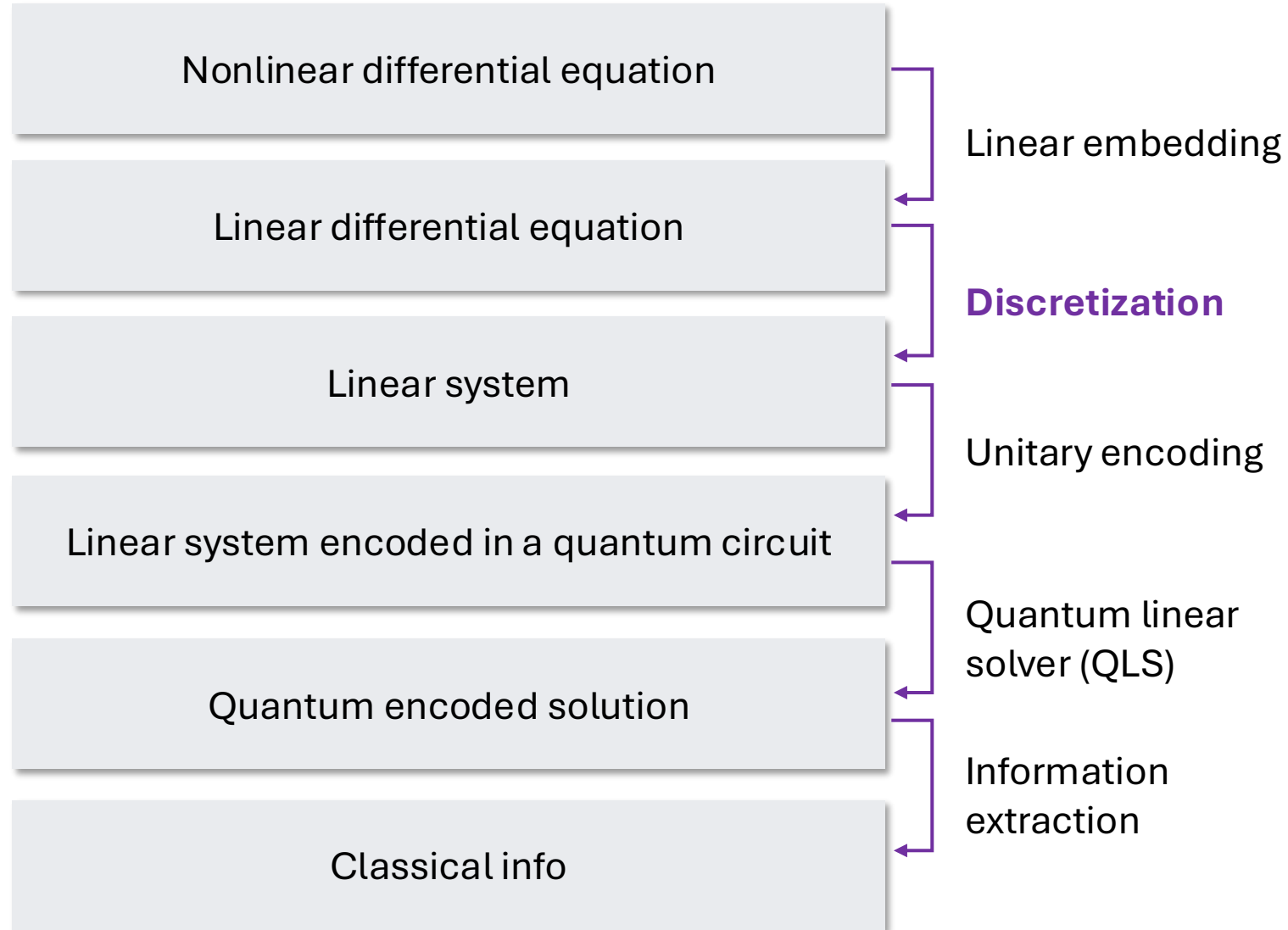


It is not guaranteed that increasing N_C reduces the embedding error ϵ_C !

What are the convergence criteria for ϵ_C , in what parameter regimes can we expect Carleman method to work?

FRAMEWORK

Complexity bottlenecks

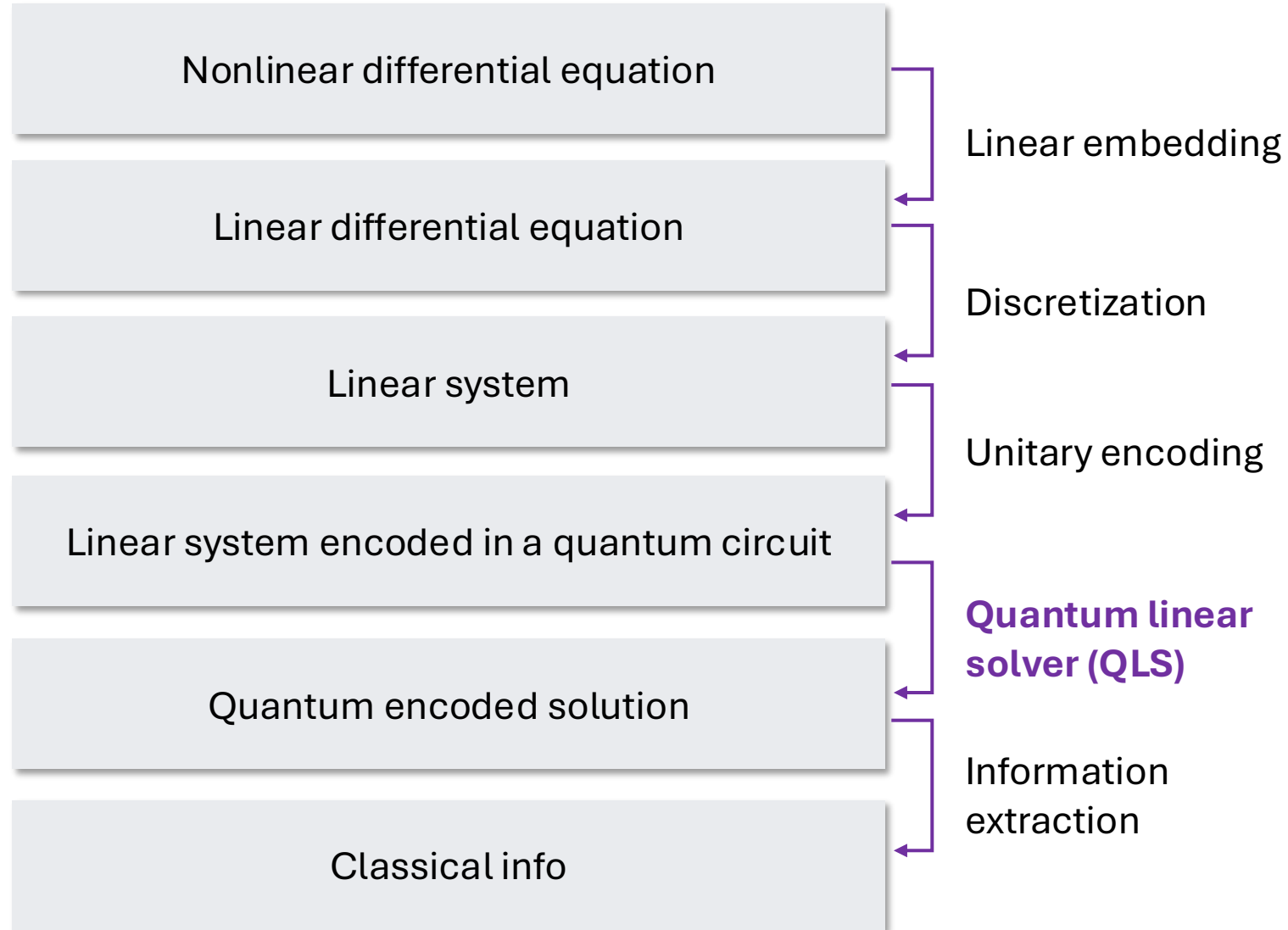


Using DBE instead of LBE yields only first order accurate method!

Prohibitively large number of time-steps needed for realistic CFD problems.

FRAMEWORK

Complexity bottlenecks

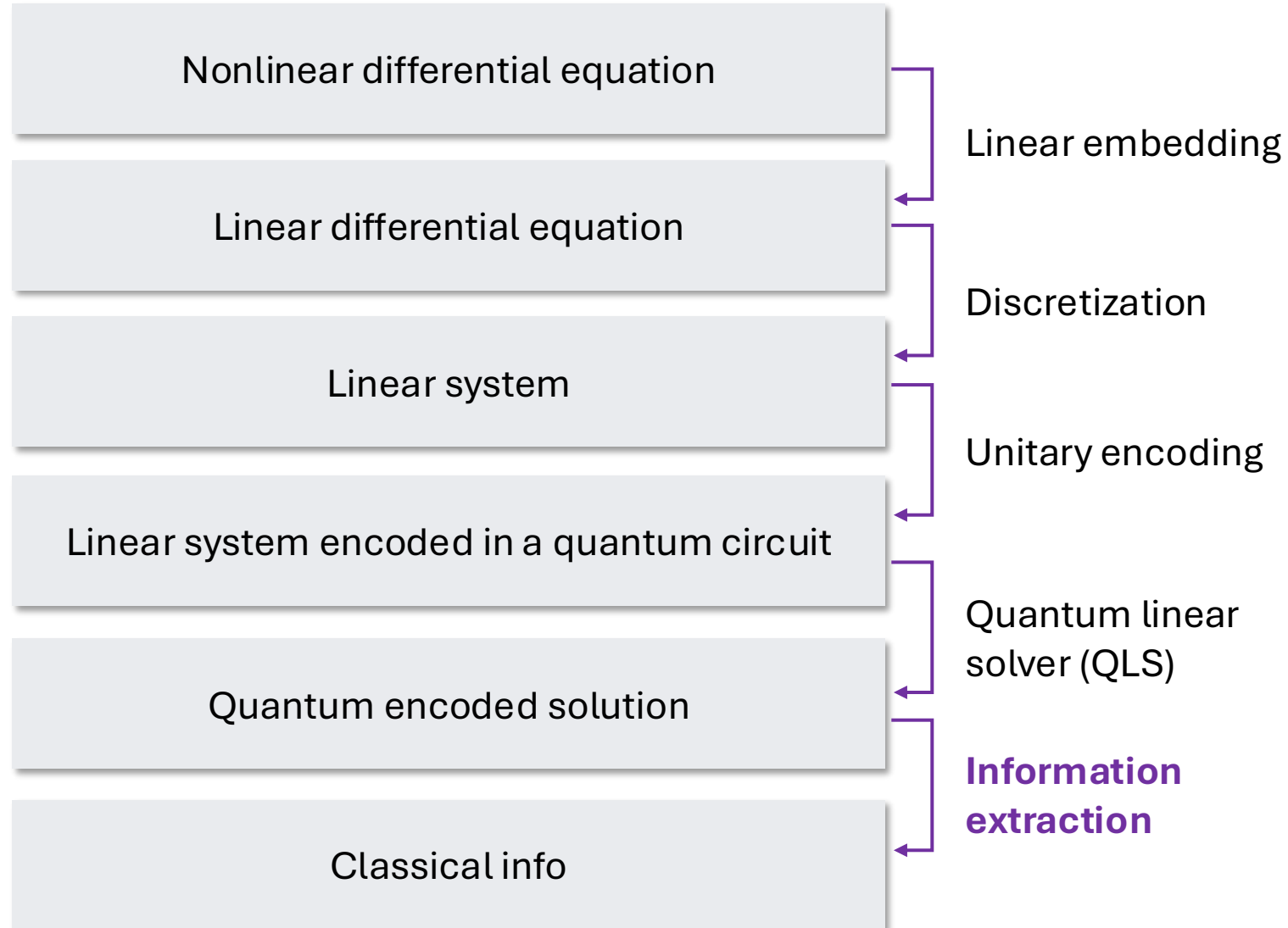


Condition number may depend polynomially on the size of the problem!

No proper analysis of the behaviour of κ in prior literature

FRAMEWORK

Complexity bottlenecks



Info extraction from the Carleman vector either inefficient or inaccurate!

Due to norm of f scaling as $\sqrt{N}$, the norm of $y = [f, f^{\otimes 2}, \dots, f^{\otimes N_c}]$ is concentrated in the last block that evolves linearly.

FRAMEWORK

Classical problem reformulation

Work with actual lattice Boltzmann equation (LBE)

$$f_m(\mathbf{r} + \mathbf{e}_m \Delta t, t + \Delta t) = f_m(\mathbf{r}, t) - \frac{\Delta t}{\tau + \Delta t/2} [f_m(\mathbf{r}, t) - f_m^{\text{eq}}(\mathbf{r}, t)]$$

Use shifted incompressible LBE & freedom of parameter selection

$$f_m \propto u^* \quad \text{and} \quad u^* = O(1/\sqrt{N}) \quad \Rightarrow \quad \|f\| = O(1)$$

Employ discrete Carleman embedding

Nonlinear discrete-time update rule can be replaced by a linear one in a larger space – we developed a novel discrete Carleman embedding for the LBE problem.

Discretization problem solved

Treated at the classical level with second order accurate method

Info extraction problem solved

The norm is concentrated in the first block

Two problems left to be analyzed:

Linear embedding accuracy
(Carleman convergence)

&

Quantum speedup
(condition number scaling)

AGENDA

0 1 Introduction

Background & summary

0 2 Framework

CFD problems & pipeline

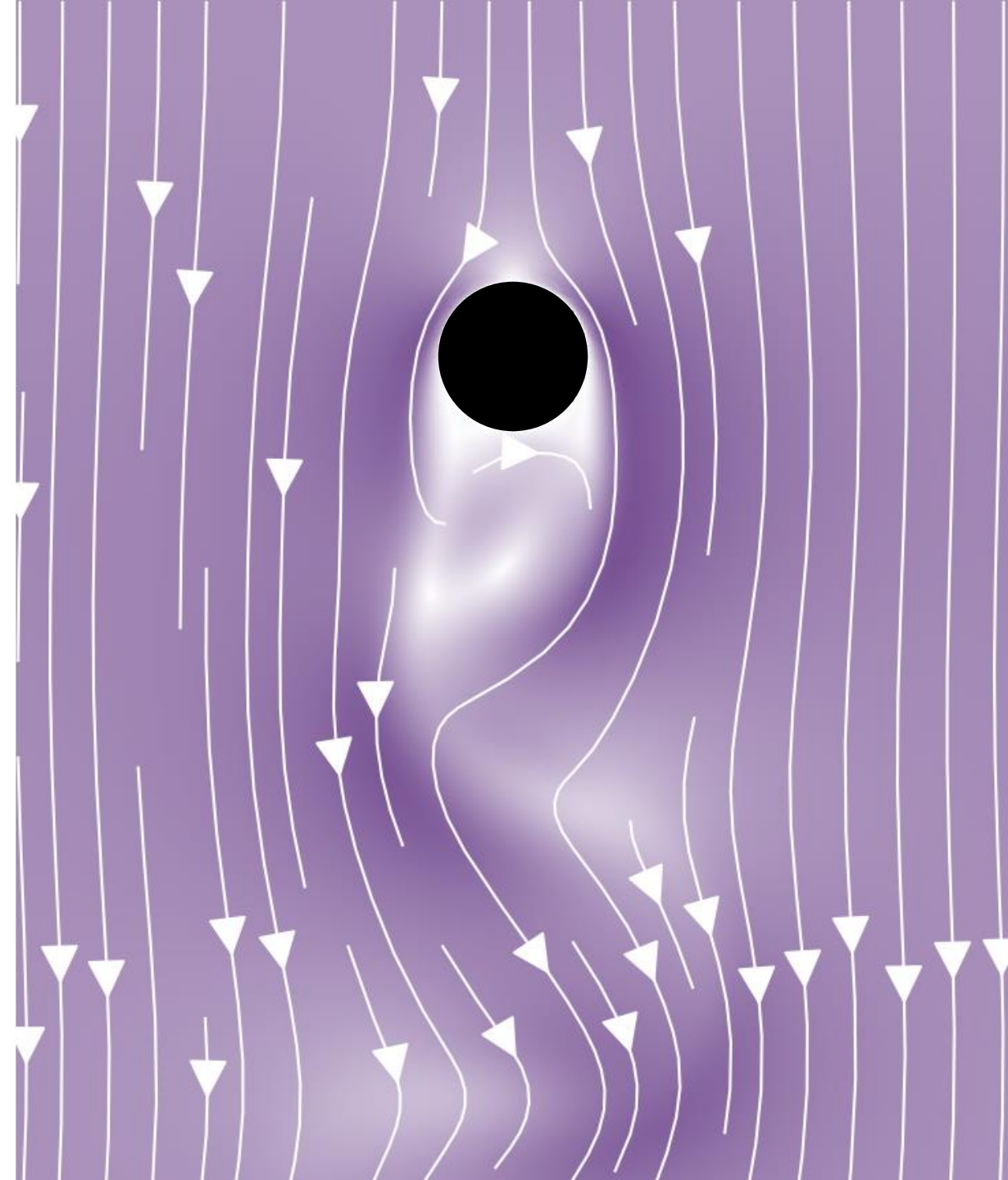
0 3 Quantum algorithm

Performance & accuracy of the quantum LBE algorithm

- Central questions & benchmark problems
- Linear embedding accuracy
- Convergence threshold
- Quantum speedup

0 4 Conclusions & outlook

Main takeaways & areas for improvement



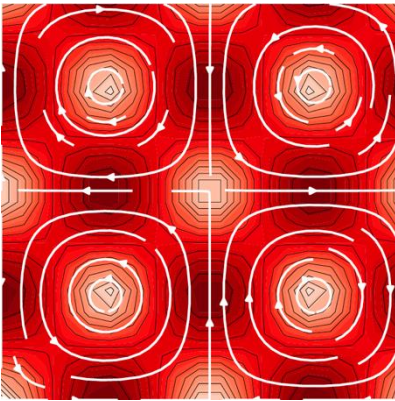
Central questions & benchmark problems

Central questions

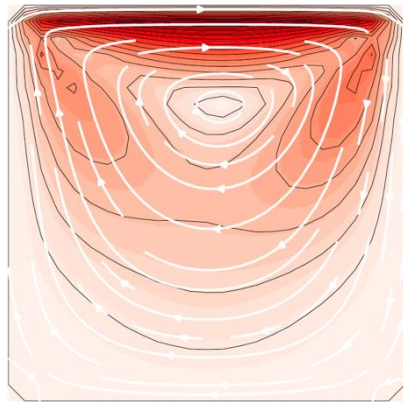
Linear embedding accuracy: how does the embedding error ϵ_c behave with growing N_c ?

Quantum speedup: how does the quantum computational complexity grow with Re ?

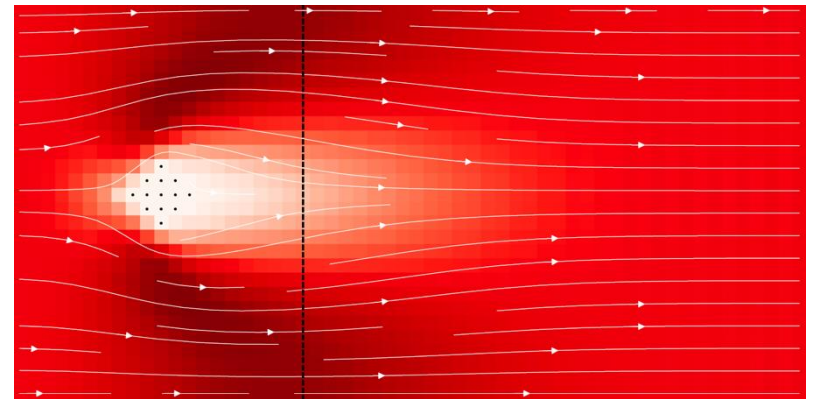
Classical benchmarking on canonical CFD problems



Driven Taylor-Green vortex



Lid-driven cavity flow



Flow past a cylinder

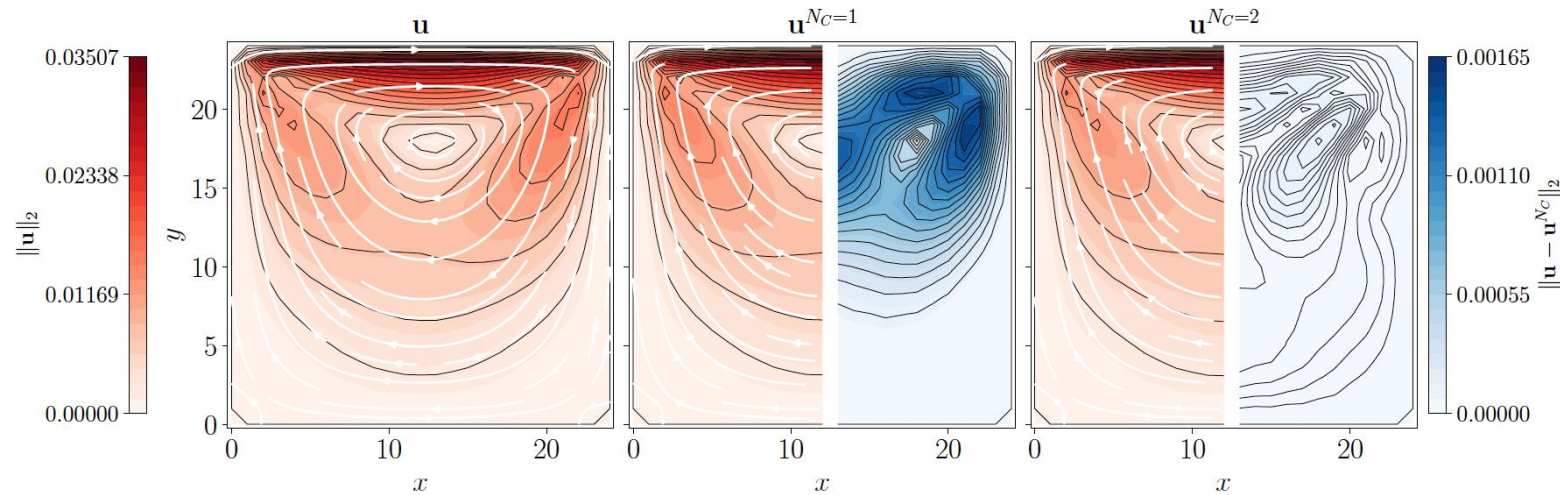
QUANTUM ALGORITHM

Linear embedding accuracy

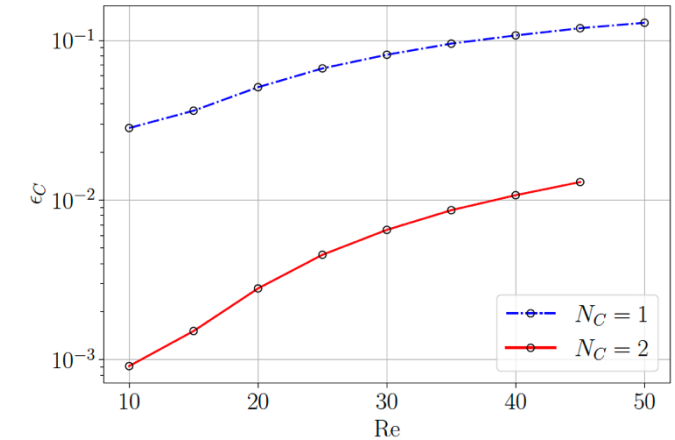
Embedding error measures

$$\epsilon_{rel}(t) := \frac{\|\mathbf{u}^{N_C}(t) - \mathbf{u}(t)\|}{\|\mathbf{u}(t)\|} \quad \epsilon_C(t) := \max_t \epsilon_{rel}(t)$$

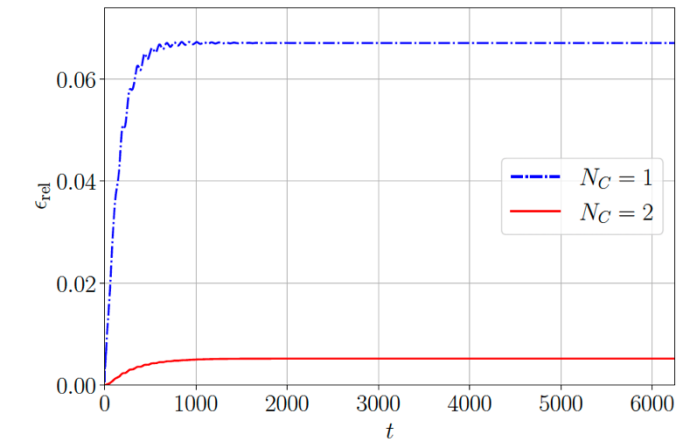
Steady state spatial embedding error for lid-driven cavity flow



Embedding error convergence

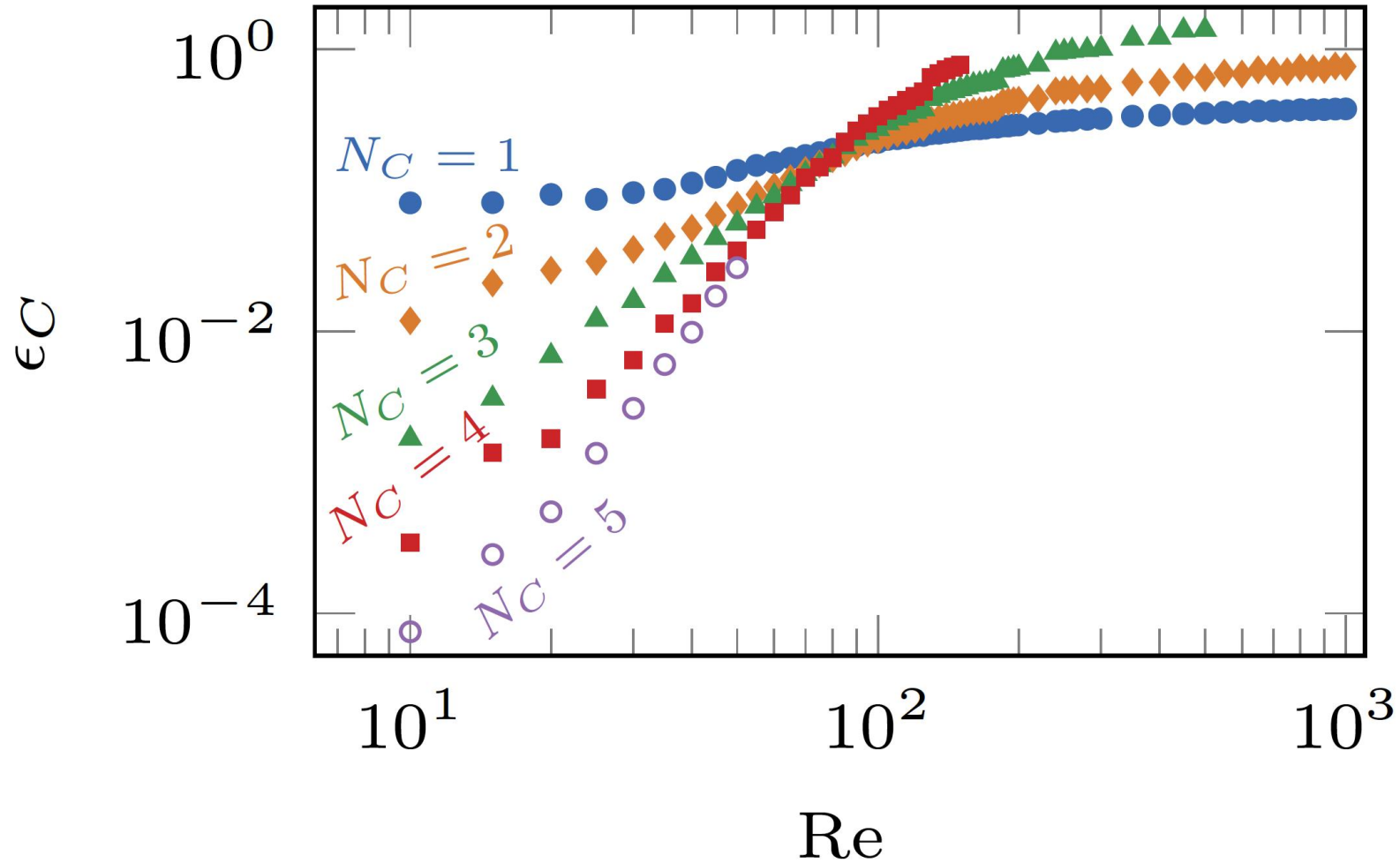


Embedding error boundedness



QUANTUM ALGORITHM

Convergence threshold



1D periodic system

Initial velocity field sinusoidal

QUANTUM ALGORITHM

Quantum speedup

Quantum complexity contributions

- Initial state preparation
- Block-encoding U_A of the linear system A
- Solving the linear system A
- Information extraction

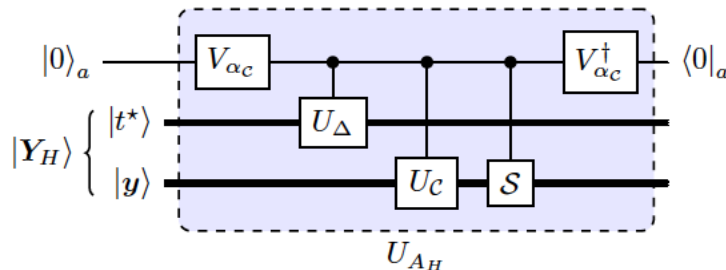
$$\text{Complexity}_Q = g \times q \times q_M$$

Number of gates
in a circuit U_A

Number of circuits
 U_A used by QLS

Measurement
info extraction cost

Compiled the core parts of the circuit to
show that g brings just a constant overhead



For drag force: $q_M = O(Re^{3/8})$

For bulk observables expecting: $q_M = O(1)$

Query complexity q is the
main complexity factor

QUANTUM ALGORITHM

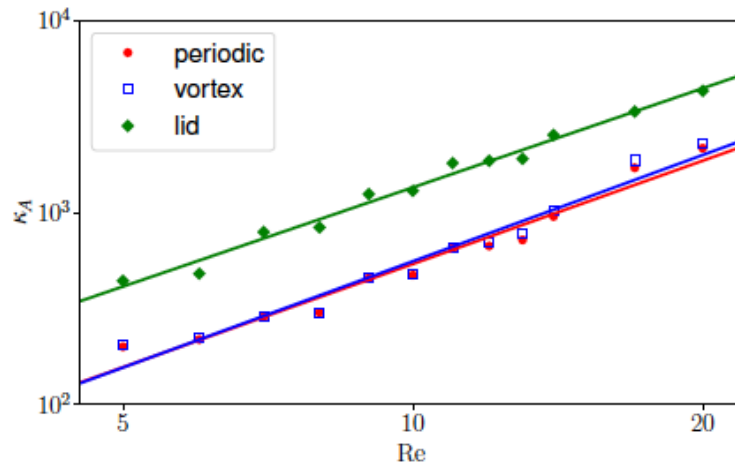
Quantum speedup

Complexity lower bound

$$\text{Complexity}_Q \geq \text{Re}^{\frac{3}{4}(1+\frac{D}{2})} \quad \text{vs} \quad \text{Complexity}_C = \text{Re}^{\frac{3}{4}(1+D)}$$

At most lower-than-quadratic speedup

Numerical complexity estimate



For 2D problem with $N_C = 2$:

$$\text{Complexity}_Q \simeq \text{Re}^{1.777} \times q_M$$

Modest quantum speedup

Bulk observables: Speedup $\simeq \text{Re}^{0.473}$

Drag force: Speedup $\simeq \text{Re}^{0.098}$

AGENDA

01 Introduction

Background & summary

02 Framework

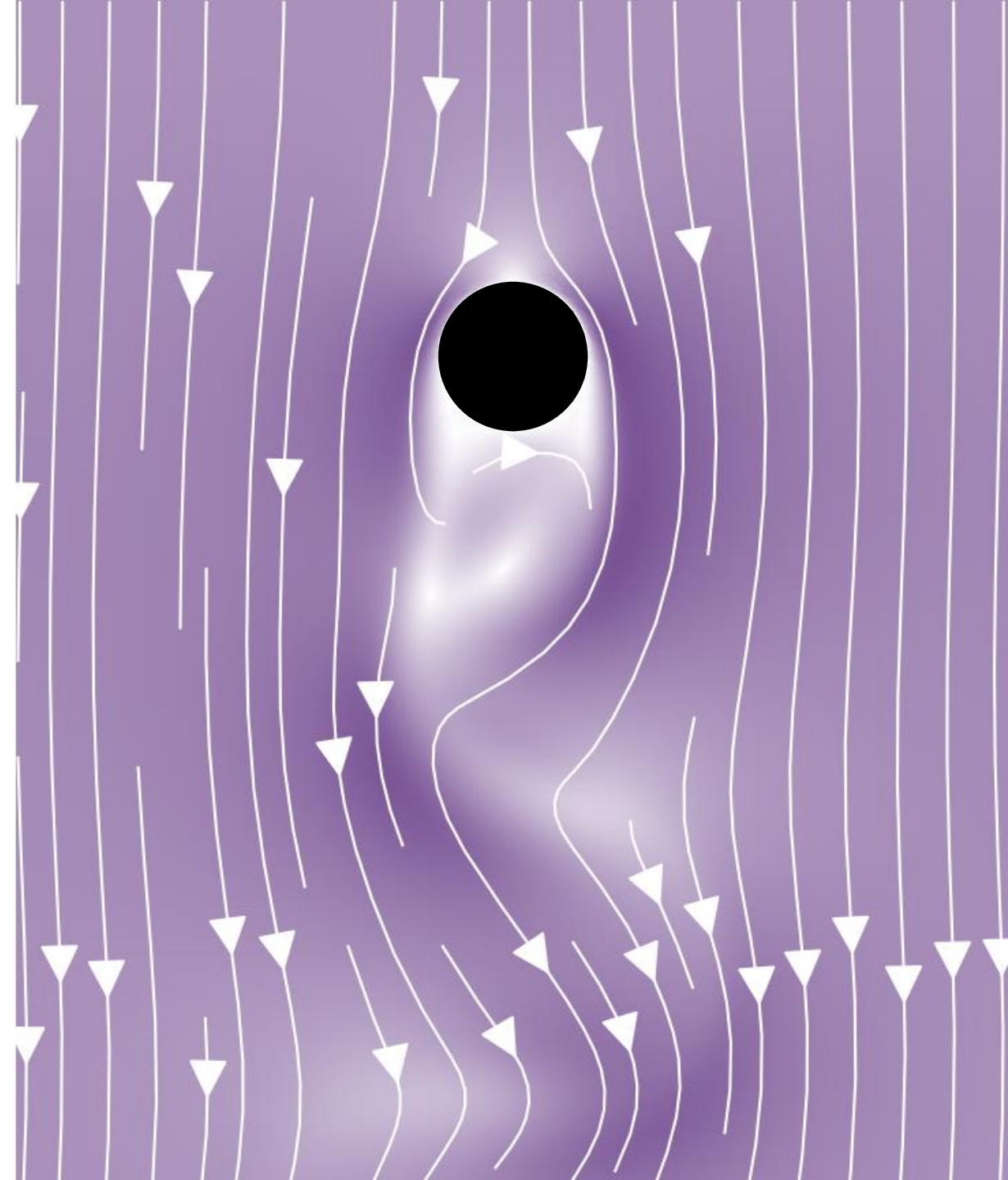
CFD problems & pipeline

03 Quantum algorithm

Performance & accuracy of the quantum LBE algorithm

04 Conclusions & outlook

Main takeaways & areas for improvement



CONCLUSIONS & OUTLOOK

Main takeaways & areas for improvement

Carleman linear embedding method converges for benchmark CFD problems

Next steps:

- Code optimization
- GPU acceleration
- Analytical convergence proofs
- Alternative linear embeddings

Modest quantum speedup for CFD problems is in principle possible

Next steps:

- Preconditioning methods
- Novel quantum linear solvers
- More efficient observables
- Spectral methods

Thank you!

